Protein structure-function relationship

In this TBL, our goal is to integrate the physiology of oxygen delivery with the biochemistry of hemoglobin. You are probably familiar with the general function of hemoglobin as an oxygen carrier. Here we will review the biochemistry of hemoglobin in a physiologically relevant context. We will review the concepts of ligand binding and allosteric regulation because it is such a fantastic example to understand how the reversible binding of a ligand to a protein can affect the physiology of the body.

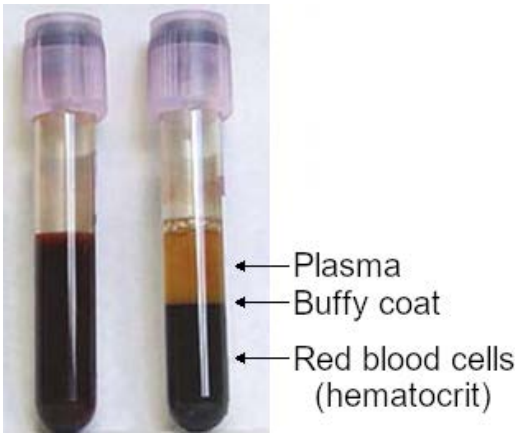
In this session we will explore two key questions. First, we will consider **why we need hemoglobin to carry oxygen**. Second, we will describe the **structure of hemoglobin and the allosteric regulation of its function**.

Concepts of oxygen content, hemoglobin saturation and hematocrit



This image shows samples of arterial blood (left) and venous blood (right). The difference in color is due to the fact that the hemoglobin molecule has a different pattern of light absorption when it is bound to oxygen. Clinically, this difference can be measured with a simple device called a **pulse oximeter**.

We can ask two questions about these samples. First, *how much oxygen is in the blood on the left compared to the blood on the right?* This is a question about **oxygen content**, or the total amount of oxygen in the sample. Oxygen content is reported in milliliters of oxygen per 100 milliliters (deciliter) of blood. The second question is, *what fraction of the hemoglobin in each sample has oxygen bound to it?* This is called **hemoglobin saturation** and is reported as a fraction or percentage. The pulse oximeter can measure hemoglobin saturation directly. If we also know the amount of hemoglobin in the blood, we can calculate the oxygen content.

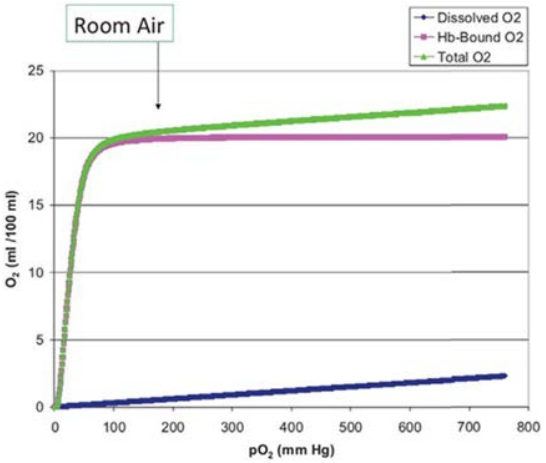


This is an image of blood collected from a venipuncture. The image shows what the blood looks like before (left) and after (right) centrifugation. When whole blood (the cellular portion together with the plasma) is placed in a tube and then spun in a centrifuge, the red blood cells are forced to the bottom of the tube because they are the heaviest element in the blood. The white blood cells and platelets are lighter, so they come to rest on top of the heavier red blood cells in a thin layer called the buffy coat. Above the buffy coat is the plasma. This technique can be used to approximate the percentage of volume that the red blood cells occupy in the total sample, which is known as the **hematocrit**. The hematocrit of a healthy adult female averages about 42, which means that 42% of the blood volume is red blood cells, while the hematocrit of a healthy adult male is about 47.

The central problem: the limited solubility of oxygen in water

The **solubility of oxygen in water is low**: only 0.3 mL of oxygen can dissolve in 100 mL of plasma. Our heart pumps 5 liters of blood per minute. Therefore, only 15 mL of dissolved oxygen can be delivered per minute by our cardiovascular system. However, our body uses about 250 mL of oxygen per minute at rest (only 6% of the total oxygen needed can be delivered in plasma! at rest!). This explains why **we need a system to transport oxygen in the blood**, i.e. **hemoglobin**. Thanks to the presence of hemoglobin, the total amount of oxygen that can be carried in 100 ml of blood is about 20 ml.

This graph shows the oxygen content at various oxygen pressures. It illustrates the large difference that exists between the ability of plasma and hemoglobin to carry oxygen at atmospheric pO_2.



Note that at room air pO_2 (160 mm Hg), dissolved oxygen is a very small contributor to the total amount of oxygen present in the blood; **the vast majority of oxygen in the blood is bound to hemoglobin** (approximately, 98%).

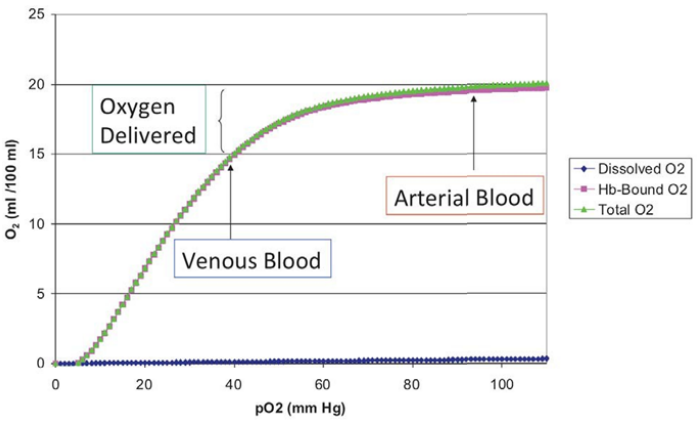
Another interesting observation is that there is a linear increase in dissolved oxygen as the pO_2 increases (blue line). However, **the oxygen content of the blood is not a linear function of the pO_2, but has a sigmoidal shape** (pink line).

In summary, the presence of hemoglobin in the blood solves the problem of oxygen delivery, because it greatly increases the amount of oxygen that can be carried in the blood.

How much oxygen can be delivered to the tissues?

As we saw before, the amount of oxygen that the blood can carry depends on the presence of hemoglobin and the pO_2. To understand the physiology of oxygen delivery, we need to know how the pO_2 of the blood changes as it circulates through the body. Blood pO_2 decreases as the blood moves from the lungs to the tissues, where the oxygen is consumed.

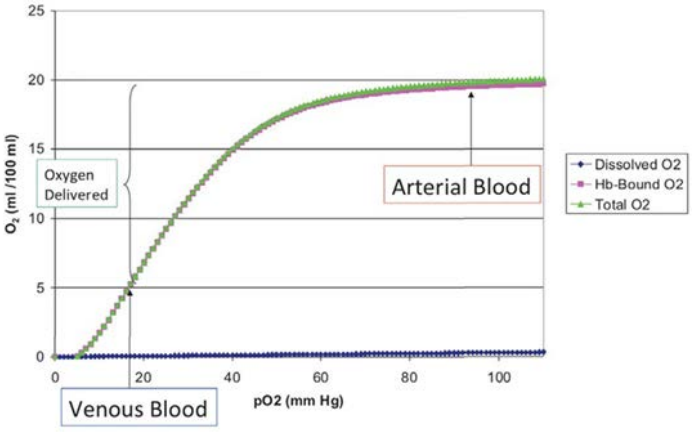
The pO_2 of inspired air is 160 mm Hg. In the lungs, the pO_2 drops to 104 mm Hg due to the presence of carbon dioxide and water vapor. Because of incomplete oxygenation of the blood by the lungs, the pO_2 of the arterial blood is approximately 95 mm Hg. Oxygenated blood circulates to the tissues, where the oxygen is released from hemoglobin. The consumption of oxygen by the tissues reduces the pO_2 of the venous blood to about 40 mm Hg for a tissue at rest (it can approach zero in an exercising muscle).



Now that we know how pO_2 differs between arterial and venous blood, we can look at the curve we discussed earlier to determine how much oxygen is being delivered to the tissues.

Arterial blood carries about 20 mL of oxygen per dL of blood, while venous blood carries about 15 mL of oxygen per dL. The difference is about 5 mL of oxygen per 100 mL of blood, which represents the oxygen consumed by the tissues. This means that **at rest, about 25% of the total oxygen carried by the blood is used by the tissues**. This percentage represents an average number. Some tissues use a much greater percentage of the total oxygen in the blood than others. In any case, doesn't that seem like a waste of capacity? *Why is the total oxygen-carrying capacity so much greater than the amount actually used?* One possible explanation is that this "excess" capacity is very helpful during exercise, when total oxygen consumption is much higher and venous pO_2 drops sharply.

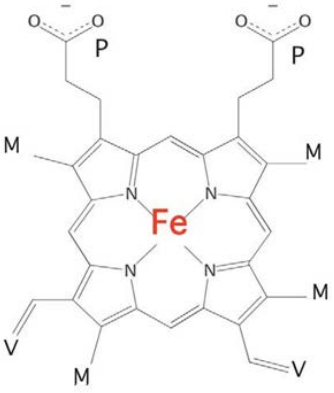
This graph shows that at a venous pO_2 of 18 mm Hg, the blood is now delivering 15 mL of oxygen per dL, a 300% increase over the 5 mL of oxygen per dL of blood delivered at rest. Note that the **sigmoidal shape** of the curve helps make this large degree of oxygen relief possible, as it provides a sharper transition as oxygen levels drop.



Finally, remember that in addition to pO_2, the other key factor affecting the total amount of oxygen that is carried by the blood is the hemoglobin concentration. Note that hemoglobin concentration is related to hematocrit. A key determinant of hematocrit is the production of red blood cells by the bone marrow, which is regulated by the hormone erythropoietin.

How does hemoglobin bind oxygen?

Oxygen binds to hemoglobin via the **heme prosthetic group**.



Heme is a metal complex of iron (Fe) and porphyrin, a type of molecule with nitrogen-containing rings. The nitrogen atoms coordinate with the iron to hold it in place. The iron is present in the Fe^{2+} oxidation state, which can coordinate with six ligands: the 4 nitrogen atoms in the rings, and two ligands that project from the iron atom perpendicular to the rings.

On one side, the iron coordinates with a **histidine** residue in hemoglobin, helping to hold the heme in place. The other side is empty, providing a binding site for the oxygen molecule.

The heme group is located in a hydrophobic cleft in the globin protein. The hydrophobic methyl (M) and vinyl (V) groups nestle into the cleft on the inside of the protein, while the hydrophilic propionic acid (P) groups protrude into the water. This gives the heme access to water, which is important because oxygen is dissolved in water.

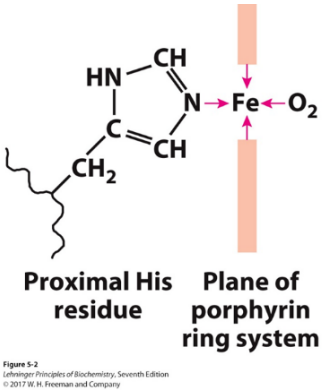


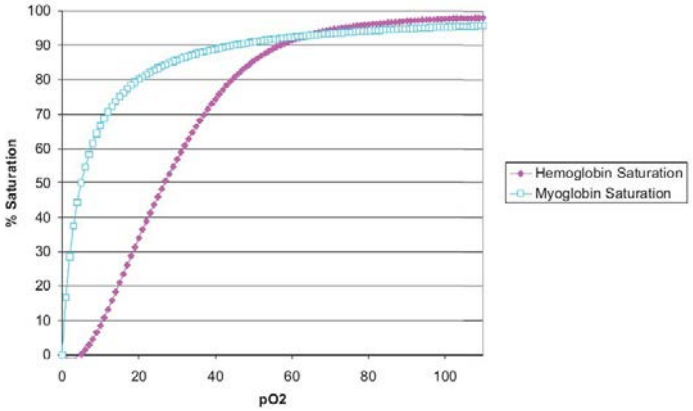
Figure 5-2
Lehninger Principles of Biochemistry, Seventh Edition
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Hemoglobin is not the only protein that contains the heme group. Heme is also found in **myoglobin**, which also binds oxygen; in **cytochromes**, which are part of the electron transport chain); and in **proteins that catalyze oxidation/reduction processes** (e.g., P450 cytochrome enzymes, which are involved in the metabolism of endogenous molecules and drugs).

Hemoglobin vs. myoglobin

Hemoglobin is found in the bloodstream, while myoglobin is found primarily in muscle tissue, where it serves as an intracellular storage site for oxygen.

Compared to hemoglobin, myoglobin has higher affinity for oxygen. This difference helps to facilitate the transfer of oxygen from hemoglobin to myoglobin. Of note, myoglobin binds oxygen less tightly than cytochrome c oxidase in the electron transport chain. Thus, there is a gradient of affinity from hemoglobin, to myoglobin, to cytochrome c oxidase, which facilitates the delivery and consumption of oxygen.

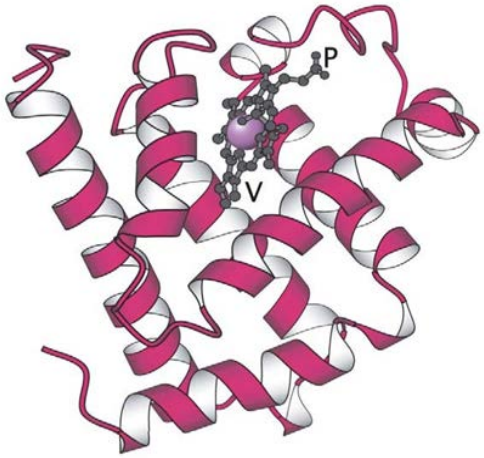


The hyperbolic shape of the binding curve for myoglobin is the typical shape for a protein-ligand interaction that exhibits saturation binding (think of the shape of the Michaelis-Menten curve for enzymes). Now check the fraction of oxygen released from myoglobin as the pO_2 drops from the arterial pO_2 of 95 mm Hg to the venous pO_2 of 40 mm Hg. Compare this to the fraction of oxygen released from hemoglobin. It should be obvious why hemoglobin is a better oxygen carrier. **The sigmoid shape of the oxyhemoglobin dissociation curve is of great physiological importance because it allows a substantial transfer of oxygen from hemoglobin to tissues with only a small decrease in pO_2**.

During the TBL we will learn at the molecular level why hemoglobin and myoglobin have different saturation curves, based on the quaternary structure of hemoglobin. We will also learn what factors facilitate oxygen release from hemoglobin and the physiological implications of this.

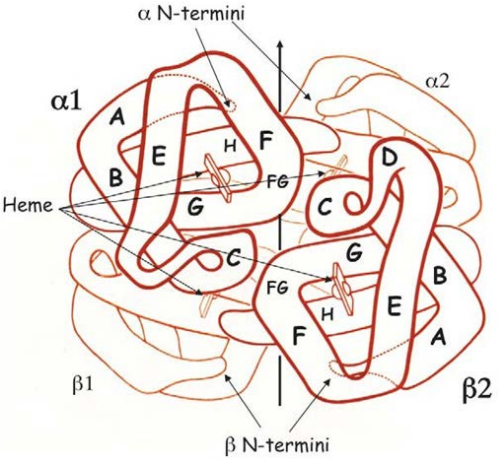
Structure of myoglobin and hemoglobin

Myoglobin is a heme-containing protein found in high concentrations in muscle cells. It is structurally related to hemoglobin and functions to facilitate the diffusion of oxygen. To understand the structure-function relationships in hemoglobin, it is helpful to first look at myoglobin (since, unlike hemoglobin, it consists of only a single subunit).



Myoglobin contains a high proportion (75%) of α-helixes in its structure. In the figure, the heme group is shown as a gray ball and stick model with the large central iron atom in purple. Note that the hydrophobic vinyl groups face the inside of the heme binding pocket and the hydrophilic propionic acid side faces away from the pocket. Heme has one covalent interaction between the heme iron and a histidine (not visualized here), and many non-covalent interactions with other residues of the protein.

Unlike myoglobin, **hemoglobin** consists of four globin subunits. Normal adult hemoglobin, or hemoglobin A, has two α subunits and two β subunits (the designation α and β is merely a naming convention and does not refer to the secondary structure). The two α subunits are identical in primary sequence, but are referred to α_1 and α_2 to distinguish them in the overall structure. The two β subunits (β_1 and β_2) are also identical. Therefore, **hemoglobulin is an $\alpha_2\beta_2$ tetramer**. The α and β subunits are similar in primary sequence, but not identical. The α chain contains 141 amino acids while the β chain contains 146 amino acids.



The four subunits have a myoglobin-like tertiary structure with 8 α-helices folding around a heme group. The helices are designated by a different letter for each helix. One of the key residues is the **histidine** at position 8 of the F helix (F8), as one of the nitrogen atoms in its side chain is covalently bonded to iron.

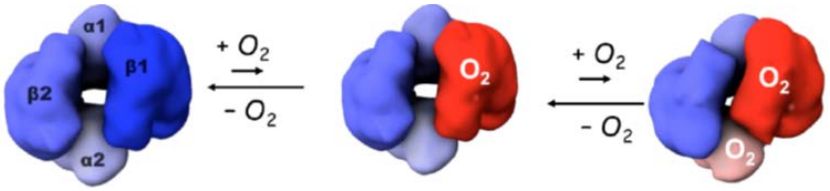
Cooperative binding in hemoglobin

Hemoglobin has a sigmoid oxygen dissociation curve. This means that hemoglobin has a relatively low affinity for oxygen at low oxygen concentrations and an increasing affinity at higher oxygen concentrations. This is an example of **cooperative binding**, in which the binding of one oxygen molecule to hemoglobin helps to bind additional oxygen molecules. This results in a "switch-like" increase in oxygen affinity. This unique feature of hemoglobin promotes **oxygen loading in the lungs** when oxygen concentrations are high, and promotes **oxygen unloading in tissues** when oxygen concentrations are low. In contrast, the hyperbolic oxygen dissociation curve we saw for myoglobin is due to a lack of cooperativity. Myoglobin is a single-subunit protein with only a single oxygen-binding site with high affinity for oxygen.

What explains the cooperative binding in hemoglobin?

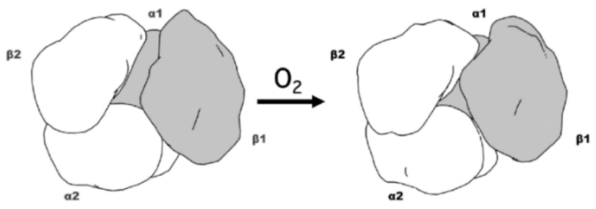
The key idea is that the hemoglobin tetramer can exist in one of two different quaternary structures: the **T structure**, which has a low affinity for oxygen, and the **R structure**, which has a high affinity (the difference in affinity is 300-fold!).

Hemoglobin is in equilibrium between the two conformers: T ↔ R. With increasing concentrations of oxygen, there is progressive binding of oxygen to each of the four heme groups, with an abrupt switch from T to R after two oxygen molecules have bound to hemoglobin. Thus, the subsequent binding of the last two oxygen molecules occurs rapidly.



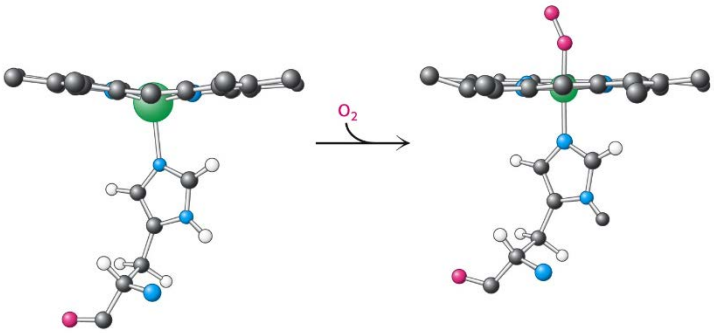
What is the difference between the quaternary structures of the T and R states?

Fully deoxygenated hemoglobin in the T (taut) state is constrained by a set of 8 ionic bonds between the different subunits, which are broken in the R (relaxed) structure when hemoglobin is fully oxygenated. The major subunit contacts are between the α and β subunits. The two α subunits barely touch each other, and the two β subunits do not touch each other at all. The α_1 subunit contacts both β_1 and β_2; similarly, α_2 contacts both β_1 and β_2. In the transition from the T structure to the R structure, the relative orientation of α_1 to β_1 does not change much, but the orientation of α_1 to β_2 changes substantially because of the breakdown of several ionic interactions. Similarly, the relative orientation of α_2 to β_2 does not change much, but the orientation of α_2 to β_1 changes. Thus, the α_1 to β_1 and α_2 to β_2 interfaces remain fixed during the T to R transition, but the α_1 to β_2 and α_2 to β_1 interfaces change during the T to R transition.



How does hemoglobin transition from the T to the R state upon oxygen binding?

Oxygen binding induces a conformational change the tertiary structure of hemoglobin.



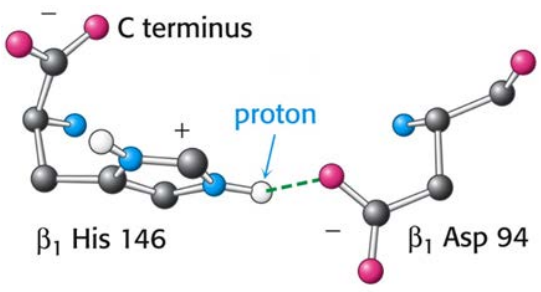
In deoxyhemoglobin, the iron atom is out of the plane of the heme group due to its interaction with histidine. When the iron is out of the ring plane, it has a low affinity for oxygen. Therefore, the T state has a low oxygen binding capacity. When the first oxygen binds to heme, the iron snaps into the plane of the heme group. This moves the histidine attached to the iron, which in turn moves the F-helix. This movement disrupts an ionic interaction present at the α_1-β_2 interface (the α_2-β_1 interface is similarly affected when oxygen binds either of these subunits). Once a second oxygen molecule binds to another heme, this results in the loss of another ionic interaction in the T state. This causes the dramatic shift to the R structure, in which all 8 ionic interactions present in the T state are lost. As these interactions are lost, the position of the remaining two F helices shifts, causing the iron to move into the plane of the heme ring. This increases the affinity of these two remaining sites for oxygen. Thus, the binding of the last two oxygen molecules occurs with much higher affinity than the first two molecules.

Allosteric regulators of the binding between hemoglobin and oxygen

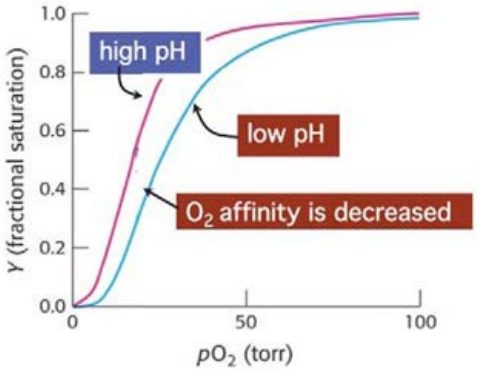
We have just learned that the presence of oxygen affects the T-R equilibrium, favoring the R state. In this sense, oxygen itself is both a ligand and a regulator of hemoglobin-oxygen affinity. We say that **oxygen acts as a homotropic allosteric effector**.

Other molecules that bind hemoglobin can also affect the T-R equilibrium if they bind more tightly to one state than to the other. Two of the most important examples of **heterotropic allosteric regulators** are **protons** and the compound **2,3-bisphosphoglycerate** (2,3-BPG; synthesized from 3-PG, a glycolysis intermediate). Both molecules bind preferentially to the T state; as opposed to oxygen. Because they stabilize the T state, protons and 2,3-BPG favor oxygen unloading.

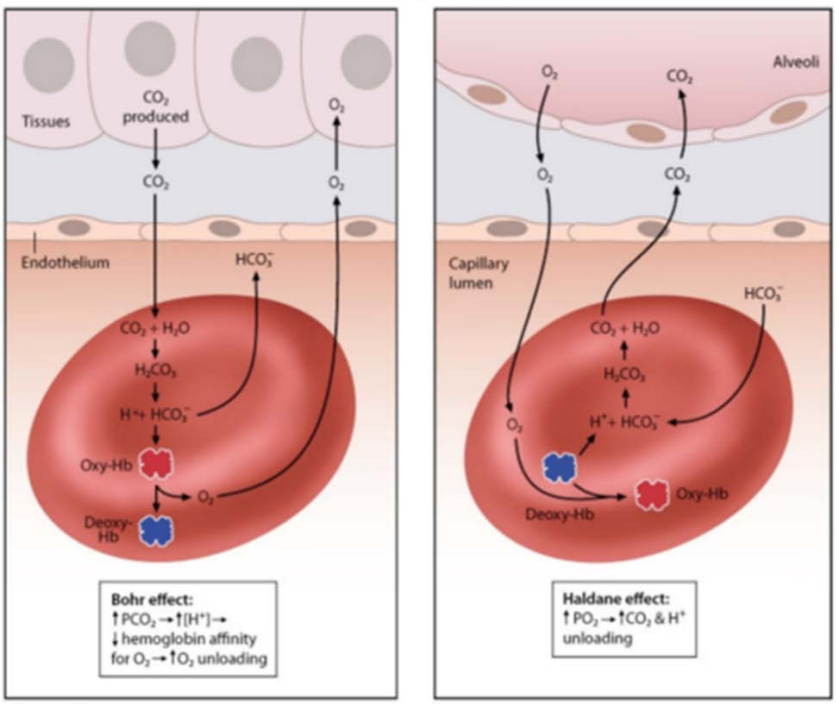
How do protons stabilize the T state?



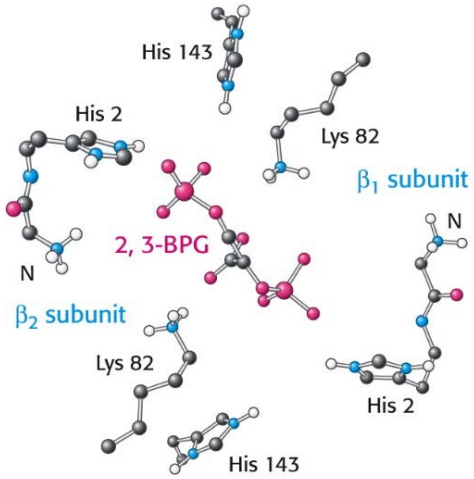
As mentioned above, several ionic interactions stabilize the T state. One of these is mediated by an interaction between the histidine at the end of the β chain (His146) and an aspartic acid in the same chain (Asp94). This ionic interaction occurs only when the histidine is protonated. As the pH increases, the histidine is deprotonated and no longer positively charged. Therefore, an increase in pH tends to favor the R state, while a decrease in pH tends to favor the T state. In other words, **a decrease in pH decreases the oxygen affinity of hemoglobin**. This effect is known as the **Bohr effect**.

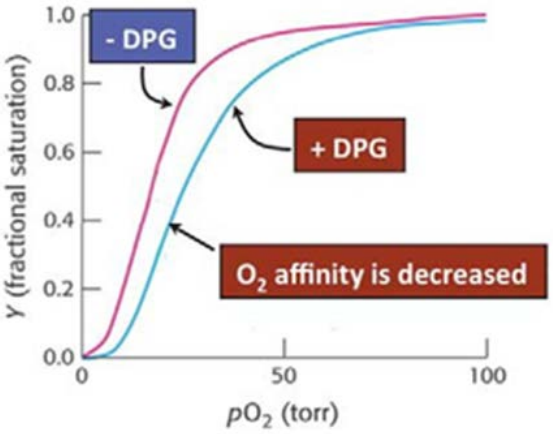


The Bohr effect is physiologically important because it facilitates oxygen delivery to tissues. **In peripheral tissues, the decrease in pH caused by carbon dioxide production reduces oxygen affinity, thereby facilitating oxygen delivery**. In addition, the Bohr effect explains why **hemoglobin is a powerful buffer, facilitating the transport of carbon dioxide to the lungs**. In peripheral tissues, carbon dioxide diffuses from respiring cells into plasma and then into erythrocytes, where the enzyme **carbonic anhydrase** catalyzes the formation of carbonic acid, which dissociates into a proton and bicarbonate (HCO_3^-). The protons bind to hemoglobin and stabilize the T-structure, ensuring maximum unloading of oxygen. Hemoglobin thus acts as a proton buffer by binding to protons and producing HCO_3^-, which is the primary form by which CO_2 is transported to the lungs. In the lungs, the process is reversed: HCO_3^- enters the red cells from the plasma. The high oxygen concentration in the lungs drives hemoglobin back into its oxygenated form, promoting the release of the proton. Because carbonic anhydrase catalyzes a reversible reaction, the enzyme catalyzes the regeneration of CO_2, which diffuses out of the red blood cells into the plasma and then crosses the capillary membrane and epithelial cells in the lungs to be released in the expired gas. The increase in CO_2 concentration due to an increase in O_2 concentration is known as the **Haldane effect**. In any case, the Bohr and Haldane effects are really the same phenomenon.



The other major allosteric regulator of hemoglobin is 2,3-BPG. This regulator also binds more tightly to the T state, favoring oxygen unloading. 2,3-BPG binds to a central cavity between the four globin subunits. The negative charges of 2,3-BPG interact with specific positively charged residues on the β-chains. The concentration of 2,3-BPG in normal erythrocytes is similar to that of hemoglobin (approximately 5 mM).

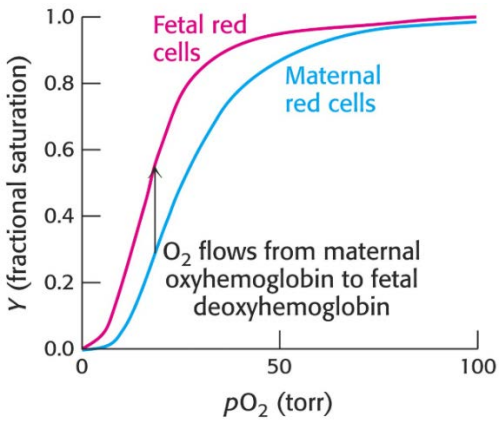




What is the physiological role of 2,3-BPG?

Under conditions of chronic oxygen deprivation (e.g., anemia or exposure to high altitude), red blood cell levels of 2,3-BPG increase, favoring increased oxygen delivery to tissues. This adaptation can take several hours to days.

Interestingly, fetal hemoglobin has γ globin chains instead of β chains. In the γ-chains, there is a substitution of a His to a Ser residue, which significantly weakens the interaction with 2,3-BPG, and thus **fetal hemoglobin has a higher affinity for oxygen**. This facilitates the delivery of oxygen to the fetus across the placenta.



Carbon monoxide poisoning

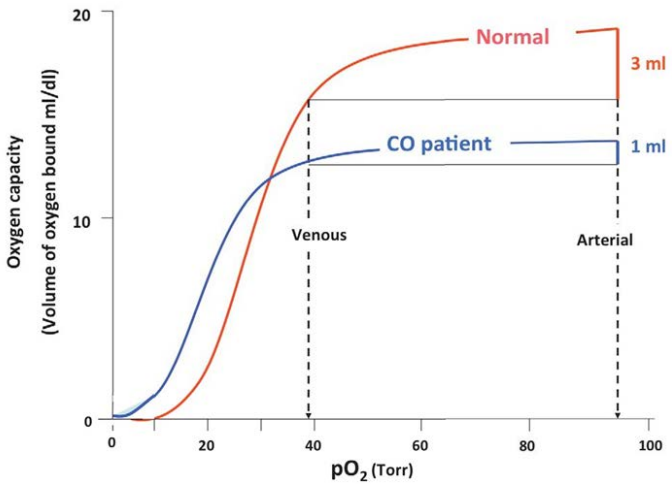
Carbon monoxide (**CO**) is a colorless, odorless gas produced as a byproduct of incomplete combustion of organic matter, especially hydrocarbons such as petroleum. It is also a major toxic component of tobacco smoke.

CO combines with hemoglobin to form **carboxyhemoglobin** (HbCO) at any or all of the oxygen-binding sites of hemoglobin. Symptoms of CO poisoning vary with the amount of HbCO. Under physiological conditions, HbCO is present at levels of less than 2% of total hemoglobin. At 6-10%, vision and time discrimination are impaired. Unconsciousness followed by death occurs at HbCO levels of 40-60% of total hemoglobin.

What is the mechanism of CO toxicity?

CO and oxygen compete for the same binding site in heme, but CO binds 210 times more tightly than oxygen. Chemically, CO binds to hemoglobin like oxygen does. It binds the iron in deoxygenated heme and moves the iron into the plane of the heme. Remember that the binding of two oxygens is sufficient to shift the hemoglobin to the R state. When one of the hemes is bound to CO, only one oxygen is needed to shift the structure from the T to the R state. Therefore, a small amount of CO can do a lot of damage by shifting the equilibrium to the R structure and increasing oxygen affinity, making it more difficult to deliver oxygen to tissues.

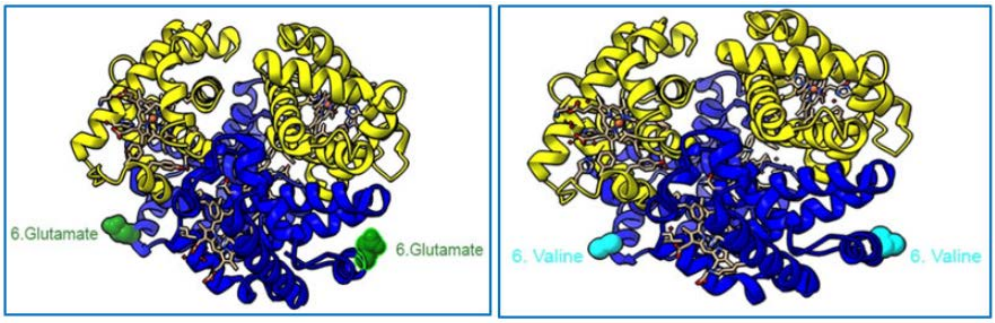
The graph shows the oxygen content curve of blood from a CO poisoning patient with an HbCO level of 25% (meaning that 25% of the oxygen binding sites are bound by CO). This effect reduces the total amount of oxygen that can be loaded onto hemoglobin. In addition, the bound CO increases the affinity of hemoglobin for oxygen, causing a leftward shift in the oxygen dissociation curve. As a result, only 1 mL of oxygen is released per 100 mL of blood (compared to 3 mL in a normal situation). Therefore, although only 25% of the oxygen binding sites are directly blocked by CO, there is a 66% decrease in the capacity to deliver oxygen.



Sickle-cell anemia

Sickle cell anemia is a vivid example of the importance of the amino acid sequence (primary structure) in determining the secondary, tertiary, and quaternary structure of a protein and, consequently, its function. Sickle cell anemia occurs in individuals who inherit two abnormal copies of the β-globin gene, one from each parent.

The alteration in β-globin consists of a **single point mutation** in the gene, located on chromosome 11, that results in the substitution of a glutamate residue for a valine residue at position 6 in the two β-chains of hemoglobin. The mutant hemoglobin is called hemoglobin S (HbS).



Normal Hemoglobin **Sickle Cell Hemoglobin**

Normal red blood cells are quite elastic and can deform to pass through capillaries. In sickle cell disease, rigid blood cells are unable to deform as they pass through narrow capillaries, leading to vascular occlusion (ischemia) and hemolysis (anemia). Healthy red blood cells typically function for 90-120 days, but sickled cells last only 10-20 days.

Interestingly, the presence of sickled red blood cells confers resistance to malaria. The malaria parasite spends part of its life cycle in red blood cells. In an HbS carrier, the premature rupture of red blood cells renders the *Plasmodium* parasite unable to reproduce. Not surprisingly, sickle cell disease is particularly prevalent in areas where malaria is still endemic.

How can a single amino acid substitution in a protein have such a dramatic effect?

The side chain of glutamate has a negative charge at pH 7.4, whereas the side chain of valine has no electrical charge. Therefore, HbS has two fewer negative charges than normal Hb (HbA). The absence of charges creates a **"sticky" hydrophobic contact point on the surface** of deoxygenated HbS. This causes HbS molecules to stick together and form long fibrous aggregates. These aggregates make red blood cells less deformable and more fragile.

